Palindromic Methods: ABBA and BM4

Common composition, spatial projection and passive energy

Explicit palindromic operations

All ABBA and BM4 presets inherit `PalindromicMethod`. For signed $d = ch$, the elementary operations are

$$A_d(u, v, t) = (u + df(t, v), v, t), \qquad B_d(u, v, t) = (u, v + df(t, u), t), \qquad T_d(u, v, t) = (u, v, t + d).$$

Only T advances time. A coefficient $c = 1/2$ means duration $h/2$; there is no additional half inside A or B. The optional exact C flow rotates the copy difference and requires an independent coupling object. Both BM4 presets omit C.

In chronological order, standard ABBA2 is $A_{h/2}, B_{h/2}, T_h, B_{h/2}, A_{h/2}$. The distinct symmetric variant $T_{h/2}, A_{h/2}, B_{h/2}, B_{h/2}, A_{h/2}, T_{h/2}$ evaluates every spatial shear at the midpoint time. Both exhibit second order on the tested nonautonomous rotation. Arbitrary palindromes do not inherit fourth or sixth order, and an arithmetic projection need not be reversible or symplectic.

`Composition` denotes an explicit tuple of operations. Consecutive time translations combine as $T_a T_b = T_{a+b}$. The generic `PalindromicImplicit` and `PalindromicMidpoint` methods select one exterior Hairer projection or a final arithmetic mean. The signed spatial flows and passive momentum normalization are retained; observers reconstruct their groups from elementary traces without replay. NumPy and JAX execute the same explicit operations and differentiate their retained shear inputs. See `docs/models/palindromic/simulation/palindromic-operations.md`.

1 Scope and public methods

The directory `src/methods/palindromic/` contains the common implementation of ABBA, BM4 and generic palindromic methods. It introduces no public method named Extended. ABBA2Implicit, ABBA4Implicit and BM4Implicit directly inherit `ProjectedPalindromicMethod`. ABBA6 has been removed; its retained theory and saved results describe historical implementations. Internal imports use the defining modules. Old ABBA and BM4 import adapters and historical `simulation.methods` routes have been removed.

For N independent planar particles the accepted physical state lies in $\mathbb{R}^{2N}$. Scalar time and optional N physical momenta are separate. Two temporary copies in $\mathbb{R}^{4N}$ are used only inside the composition and projection; they are discarded before acceptance.

2 Direct and adjoint compositions

Let χ_s denote a first-order map on the extended spatial workspace, with adjoint $\chi_s^* = \chi_{-s}^{-1}$, including the appropriate time translation for nonautonomous dynamics. A recipe with $2r$ stages is

$$\Psi_h = \chi_{b_{2r}h} \circ \chi^*_{b_{2r-1}h} \circ \cdots \circ \chi_{b_2 h} \circ \chi^*_{b_1 h}.$$

In execution order the stages alternate adjoint, direct. The weights obey $b_j = b_{2r+1-j}$ and $\sum_j b_j = 1$. Reflection exchanges each map with its adjoint. The recipes explicitly list the elementary operations and combine consecutive time translations. Run preparation checks reflected operation pairs and unit A, B and T totals; higher-order conditions remain necessary.

With stage clock τ and signed duration $s = b_j h$, the adjoint evaluates at τ, the direct map at $\tau + s$, and the clock subsequently becomes $\tau + s$. Negative coefficients reverse the clock as well as the

subflow. For uncoupled GC maps, the pair with weights $(1/2, 1/2)$ reproduces the endpoint-time A–B–B–A construction. Each direct or adjoint map contains two spatial shears. Both BM4 variants also omit harmonic coupling. The exact C flow remains independently available through `HarmonicCoupling`, outside BM4.

ABBA4 uses three unprojected ABBA pairs scaled by

$$\gamma = \frac{1}{2 - 2^{1/3}}, \qquad \delta = -\frac{2^{1/3}}{2 - 2^{1/3}}, \qquad (\gamma, \delta, \gamma).$$

Its six direct/adjoint weights are $(\gamma/2, \gamma/2, \delta/2, \delta/2, \gamma/2, \gamma/2)$. BM4 retains its twelve mirrored weights and its own fourth-order recipe. ABBA4 declares the complete operation tuple, just as ABBA2 does. Its single projection surrounds all three unprojected pairs; it is not a sequence of three separately projected ABBA2 steps.

3 One common spatial projection equation

Write $m = 2N$, $Ez = (z, z)$, $G = [I, -I]$ and $N_0 = G^\mathsf{T}$. The reduced Hairer projection finds $\mu \in \mathbb{R}^m$ such that

$$r(\mu) = G\Psi_h(Ez + N_0\mu) + 2\mu = 0.$$

After convergence the two corrected outputs are averaged. With $P = \frac{1}{2}[I, I]$, the accepted state is

$$z_+ = P\big(\Psi_h(Ez + N_0\mu) + N_0\mu\big).$$

The same equation applies to ABBA2, the complete ABBA4 composition, and BM4; only Ψ_h changes. All have one nonlinear solve per complete step.

The simultaneous ABBA formulation solves for $(u_+, v_+, \mu) \in \mathbb{R}^{3m}$:

$$\begin{pmatrix} u_+ - \mu - [\Psi_h(Ez + N_0\mu)]_u \\ v_+ + \mu - [\Psi_h(Ez + N_0\mu)]_v \\ u_+ - v_+ \end{pmatrix} = 0.$$

Arithmetic midpoint variants instead start from Ez, apply the complete recipe, and average once. They do not solve a nonlinear equation; arithmetic projection does not imply exact symplecticity or reversibility.

4 Nonlinear solution and exact tangents

All spatial roots use the unchanged stopping threshold

$$\mathrm{atol} + \mathrm{rtol}\max(1, \|z\|_\infty).$$

The equation classes `ReducedProjection` and `SimultaneousProjection` in `src/methods/projection.py` construct these residuals and their Jacobians. The numerical method supplies its composition; the algebraic solver receives the resulting system. Newton and good Broyden share the convergence machinery in `src/solvers/nonlinear.py`. Broyden's initial reduced Jacobian is $4I$, the exact zero-step value of Dr. The simultaneous zero-step block matrix is

$$\begin{pmatrix} I_{2m} & -2N_0 \\ G & 0 \end{pmatrix}.$$

Broyden subsequently uses residual secants and evaluates no analytic Jacobian.

If $M = D\Psi_h$, Newton's reduced Jacobian is $GMN_0 + 2I$. The common composition retains the input to each spatial shear. Exact stage tangents are evaluated at those inputs and multiplied in execution order, $M = M_{2r} \cdots M_1$, without replaying vector-field stages. Independence of the particles gives 4×4 blocks for M, 2×2 reduced Newton blocks, and 6×6 simultaneous blocks. BM4 also retains its optional dense centered-difference differentiation of the complete map.

5 Historical generalized energy from accepted stages

Specialized historical studies supply $H(t,z)$ and $-\partial_t H(t,z)$ to a stage-energy observer. The generic EnergyObserver uses endpoint observations instead. The method supplies only numerical quadrature data. For the accepted spatial trace, let (t_j, s_j, z_j) denote the evaluation time, signed duration and physical source state of each individual shear. The shared quadrature is

$$\Delta\kappa = \frac{1}{2}\sum_j s_j\,[-\partial_t H(t_j, z_j)].$$

There are four shear points for ABBA2, twelve for ABBA4 and 24 for BM4. The factor $1/2$ converts summed doubled momentum to the physical variable κ. Spatial harmonic coupling does not change this passive accumulator. Energy derivatives are evaluated only after the spatial root converges, from its retained trace; trial roots do not evaluate energy. The accepted diagnostic is

$$H(t_n, z_n) + \kappa_n - H(t_0, z_0), \qquad \kappa_0 = 0.$$

This is an energy-balance diagnostic, not a claim that the time-dependent physical Hamiltonian is conserved. Tracking does not modify spatial states, nonlinear unknown dimensions or stopping scales.

6 Physical state and endpoint energy observations

Accepted states contain one physical vector $Z \in \mathbb{R}^{dN}$ and scalar time. GC vectors have $2N$ entries and FC vectors have $4N$; doubled workspaces are transient. InitialConditions supplies values and layout during preparation only. The numerical base retains functions, metadata, run status and copy count; execution owns the initial snapshot and output schedule. Supplied Jacobians are independent particle blocks (N, d, d); finite differences use dense matrices.

The generic `EnergyObserver` evaluates only

$$H_n = H(t_n, Z_n), \quad H_{n+1} = H(t_{n+1}, Z_{n+1}), \quad (\partial_t H)_n, \quad (\partial_t H)_{n+1}, \quad \Delta H_n = H_{n+1} - H_n.$$

It also evaluates these physical functions at requested saved samples, without consuming internal stages or integrating a momentum coordinate. The derivative is positive, analytic and partial at fixed state. GC differentiates its normalized Hamiltonian potential with `dt=1`; spatial splines and Fourier time factors retain their original scaling. This is not the secant $\Delta H_n/h$. Variation of a time-dependent Hamiltonian is not a numerical conservation error.

Specialized historical generalized-energy studies separately reconstruct $\dot\kappa = -\partial_t H$ from retained numerical stages. Their signed weights, physical normalization and archived diagnostics keep their original meaning. They are not part of generic energy observation. Host and JAX execution share the fixed numerical map and physical-only accepted state. JAX evaluates endpoint energy on the device without a momentum accumulator.

7 Execution and organization

Execution injects the vector field and its optional Jacobian into a fresh run; no method retains the dynamics object. Initialization binds the method's recipe, `DoubleFormulation` and configured algebraic solver once. Solver controls include tolerances, iteration limits, differentiation and iteration callbacks. The generic energy observer records endpoint values and analytic positive time derivatives. Historical stage diagnostics separately own momentum and its sampled history. The ordinary implicit execution is

`advance` $\longrightarrow$ `solve_projection` $\longrightarrow$ `compose` $\longrightarrow$ `A/B/T/C`.

ABBA2Implicit, ABBA4Implicit and BM4Implicit share direct preparation and advancement through `ProjectedPalindromicMethod`; no intermediate ABBA class remains. Midpoint variants share `MidpointPalindromicMethod`. Each recipe selects its elementary operations. Each public configuration has its own module in `src/methods/palindromic/`, such as `abba4_implicit.py` or `bm4_implicit.py`. Generic recipes use `implicit.py` and `midpoint.py`. Shared bases live in `base.py`, `projected_base.py`, and `midpoint_base.py`. Common numerical operations live in `core/`, and optional event conversion lives in `observations.py`. The general integration coordinator remains responsible for main steps, saved times, shadow output samples, collection and observer dispatch. Observers receive independent snapshots in the existing public event types; BM4 events reuse accepted traces without repeating the spatial composition.

Current diagrams and module responsibilities are documented in `docs/models/palindromic/simulation/palindromic-simulation-architecture.md`. Tests verify public identity, coefficient and projection controls, signed nonautonomous maps, exact tangents and passive tracking. Tests also check direct common inheritance and the absence of retired ABBA6 interfaces. The five remaining presets preserve their numerical maps and observations. Historical ABBA6 results retain their original method and projection labels.